



Society of Petroleum Engineers

SPE-229040-MS

Ruleset-Driven Automated Design of PAD/Platform Multi-Wells Using Cloud-Based Software: A Modern Approach to Well Engineering

P. Mwansa, Adnoc Onshore, Abu Dhabi, United Arab Emirates; M. Ali HASSAN and J. Strand, Oliasoft AS, Oslo, Norway

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229040-MS](https://doi.org/10.2118/229040-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Objectives/Scope: The adoption of PAD/Platform drilling has enabled operators to efficiently drill clusters of wells while minimizing surface footprint and infrastructure costs. However, conventional workflows for planning multiple platform wells remain time & labor-intensive, relying on legacy software and repetitive manual inputs. This paper presents the development and validation of a cloud-based, rule-set driven workflow for automating the design of PAD/Platform Multi-wells, reducing engineering time and improving consistency. The work presented in this paper is the result of a collaboration between a major Middle Eastern National Oil Company (NOC) and a Norwegian software vendor

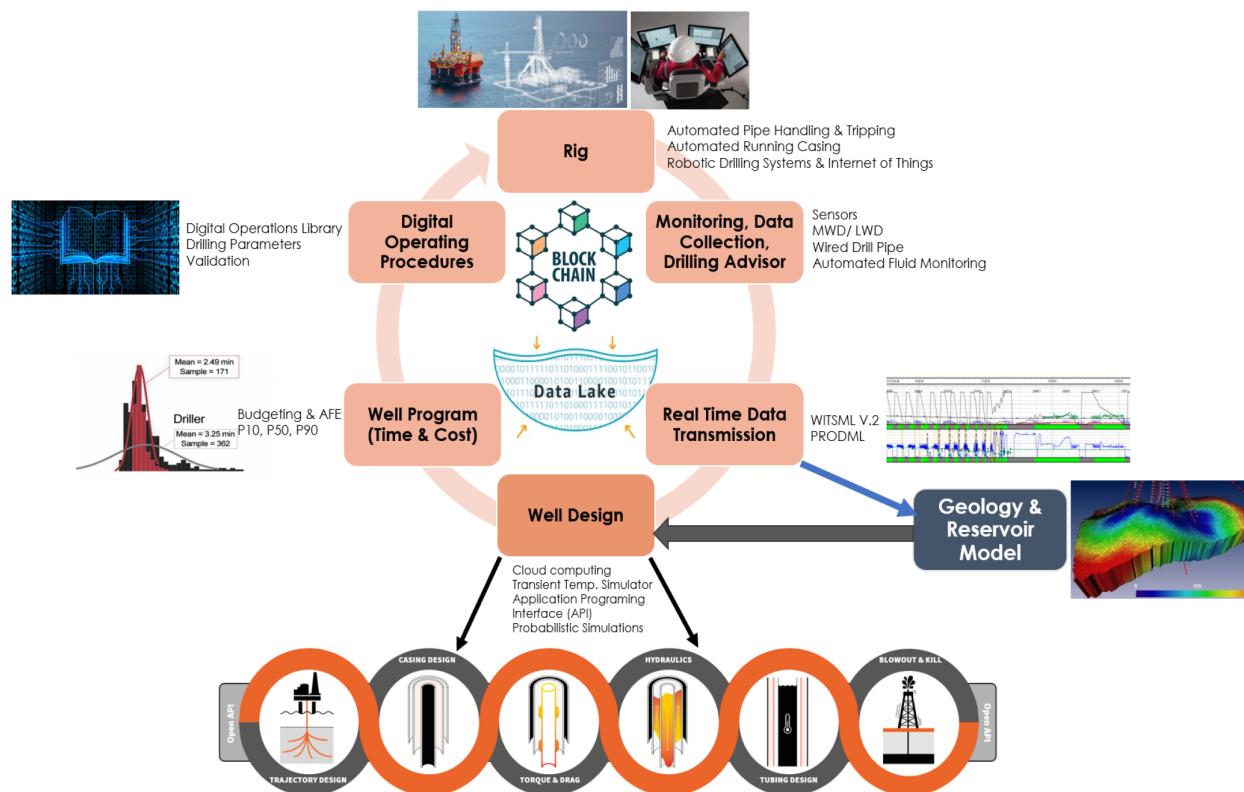
Methods, Procedures, Process: A design workflow was developed using a modern cloud-based software, named WellDesign. A comprehensive set of predefined engineering rules and simulation parameters were configured within the software, encompassing the full well lifecycle, including:

- Automatic generation of multiple trajectories & Anti-collision check
- Casing and Tubing rule-set automated design
- Drilling Mechanics including Torque & Drag and Hydraulics modeling
- Transient Temperature simulation for drilling & production operations
- Blowout risk analysis and Relief Well dynamic kill simulations

Results, Observations, Conclusions: The software's modular architecture allows seamless data transfer across all design stages. Interconnected calculation engines ensure automated, iterative computations without the need for redundant manual data entry or validation, enabling real-time updates to designs based on rule logic. For any new PAD/Platform development, the user can initiate the automated workflow via rule-based templates. The software autonomously executes the full suite of simulations and generates engineering designs across all wells in the pad. This approach achieved significant reductions in design cycle time while improving quality control and standardization. This approach demonstrates the viability

Introduction

Well Construction Automation is gradually becoming more prominent in oil & gas industry, see [Figure 1](#). It encompasses the application of digital technology in all aspects of well drilling and completion (i.e., automatic well design, digitalization of downhole tools & surface equipment, remote monitoring, real time data transmission, and robotic rig systems).



In today's Industry 4.0 transformation world, easy-to-use cloud-based software can partially automate Oil & Gas wells design. Future software development should be made to produce fully autonomous Oil & Gas well designs while maintaining industry safety & well integrity standards

The PAD Multi-well Automatic Design workflow presented in this paper is a starting step to achieve Well Construction Autonomy. This workflow made possible through modern "API" based drilling calculation engines hosted on a cloud computing environment. In the Drilling Digitalization Infrastructure, everything will be connected via Open APIs, and Real time Machine-to- Machine Communication. Modern Cloud-Based Applications will enable Engineers, within seconds, to automatically execute complex drilling simulations and calculations. The software tool used in this workflow, named WellDesign, is a cloud based single page application utilizing networks and cloud computers for data storage and collaboration. It offers a set of application programming interfaces (API), enabling machine-to-machine communication & 100% automation, where whole or parts of the software can be operated by other computers. On top of this interface, the software tool has a 2D and 3D human graphical user interface, utilizing web technologies such as accelerated 2D (Canvas, 3D.js) and 3D (WebGL, three.js) graphics. This software tool aims to run

on all kinds of computers (Windows, Macs) and other interactive devices supporting modern web browsers (tablets, phones and similar).

The following drilling engineering calculation modules are available in the subject cloud-based Well Design tool:

1. **Trajectory Design:** Complete the Wellpath modeling tool and well archive. Includes several well planning methods, Full AC Analysis, and survey data management (with QAQC capabilities). Relief well planning templates.
2. **Casing Design:** Advanced triaxial stress analysis. Integrated with advanced temperature/production simulations and AFE/APB calculations. Rule-based automated casing design.
3. **Torque & Drag:** Analysis for drilling, casing, and completion operations and assessment of the impact of Torque & Drag loads in soft, semi-stiff, and stiff strings. Customize operations, plots, and sensitivity analysis.
4. **Hydraulics:** Advanced calculation of pressure losses through the entire circulating system. Full Surge & Swab functionality. Extensive rheology profiles for several well configurations.
5. **Fluid Train:** Cementing Hydraulics simulations, Casing & Liner cement job and Cement Plug.
6. **Tubing Design:** Advanced thermal modeling, multi-string analysis, and custom operations for any well configuration. Axial and triaxial representation of all temperature-related loads.
7. **Blowout & Kill:** Probabilistic scenarios for blowout simulations with uncertainties on input parameters. Worse-case charge calculations. Dynamic kill calculations in accordance with all industry guidelines.

Benefits of Cloud Computing

IBM outlines three main benefits of cloud computing: Flexibility, Efficiency, and Strategic value."

- **Flexibility**
 - Scalability: Cloud infrastructure scales on demand to support fluctuating workloads.
 - Storage options: Users can choose public, private, or hybrid storage offerings, depending on security needs and other considerations.
 - Control choices: Organizations can determine their level of control with as-a-service options. These include software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service (IaaS).
 - Tool selection: Users can select from a menu of prebuilt tools and features to build a solution that fits their specific needs.
 - Security features: Virtual private cloud, encryption, and API keys help keep data secure.
- **Efficiency**
 - Accessibility: Cloud-based applications and data are accessible from virtually any internet connected device.
 - Data security: Hardware failures do not result in data loss because of networked backups.
 - Savings on equipment: Cloud computing uses remote resources, saving organizations the cost of servers and other equipment.
- **Strategic value**
 - Streamlined work: Cloud service providers (CSPs) manage underlying infrastructure, enabling organizations to focus on application development and other priorities.

- Regular updates: Service providers regularly update offerings to give users the most up-to-date technology.
 - Collaboration: Worldwide access means teams can collaborate from widespread locations.
 - Competitive edge: Organizations can move more nimbly than competitors who must devote IT resources to managing infrastructure.
- " (IBM, 2018)

Below is a comparison between Modern Cloud-Native Software versus Traditional Monolithic (Desktop Software)

Table 1—Modern Cloud-Native Software versus Traditional Monolithic (Desktop Software)

Aspect	Modern Cloud-Native Software	Traditional Monolithic (Desktop Software)
Architecture Style	Distributed microservices; loosely coupled services	Tightly coupled components (User interface, business logic, data access), all-in-one large unit application
Deployment	Runs in containers or serverless environments across the cloud	Runs on a single machine (desktop or server)
Scalability	Highly scalable, individual services scale independently	Limited scalability, must scale the whole application
Resilience	Fault-tolerant, failure in one service doesn't bring down the system	Fragile, a single failure can crash the entire application
Maintainability	Easier to update, services can be upgraded independently	Harder to update, requires redeploying the entire application
Technology Stack	Polyglot, different services can use different languages & frameworks	Usually a single language & framework
Digital Transformation	Strong enabler, supports agility & rapid innovation	Weak enabler, slow release cycles, less flexibility

Technology Stack

Below a brief description of the technology stack (see [Figure 2](#)) behind the cloud-based Well Design software tool presented in this paper:

- Core server technology is being served by Node.js, which act as the integration layer between the various core technologies and services in use. Data storage is handled through the PostgreSQL database, using a mixed model architecture (relational + object based). The application server runs on Linux, macOS and Windows.
- Communication (network traffic) between servers and client machines are protected with strong SSL encryption. Login passwords are not stored anywhere. Secret session keys (to store server session data client side) uses encrypted JSON Web Tokens (unreadable to the client).
- Data is stored in a PostgreSQL database. This database is not "exposed" directly to the network (external world) in any way. Only access through the server APIs is possible. Database access is only done using parameterized queries, meaning the service is immune to classical SQL injection attacks.
- Software tool uses various simulation engines. Some are developed in-house, others are provided by external parties. Simple simulations are done in javascript directly. These may be run both server side (through APIs) or client side (on browsers). Other simulations require access to compiled software or special platforms. These always have interfaces through our server and can be accessed using industry standard REST based JSON interfaces.

- Computational Fluid Dynamics (CFD) temperature simulation engine is written in high performance portable C++. It compiles and runs on Linux and macOS (and can run on Windows as well).
- Blowout simulation engine is written in portable C# code and currently runs on both Windows and Linux images.
- Specific platform requirements for these engines are not something that users see or experience; everything is wrapped and served through REST based JSON APIs application programming interfaces, see [Figure 3](#) for Graphical User Interface, there is no need for any proprietary cloud APIs. This means that services can run on any cloud.

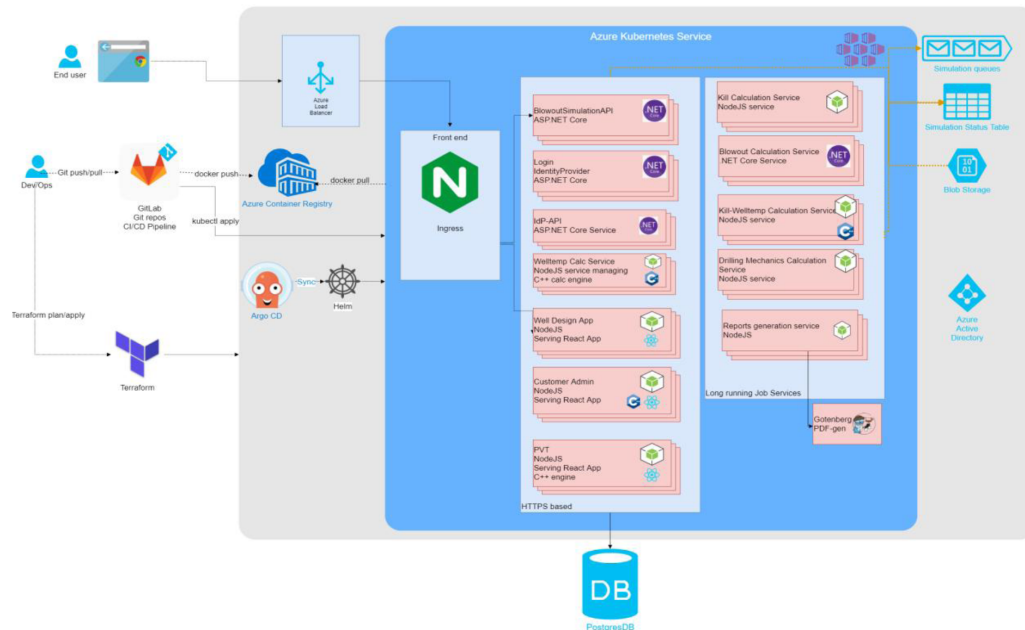


Figure 2—Software Architecture diagram

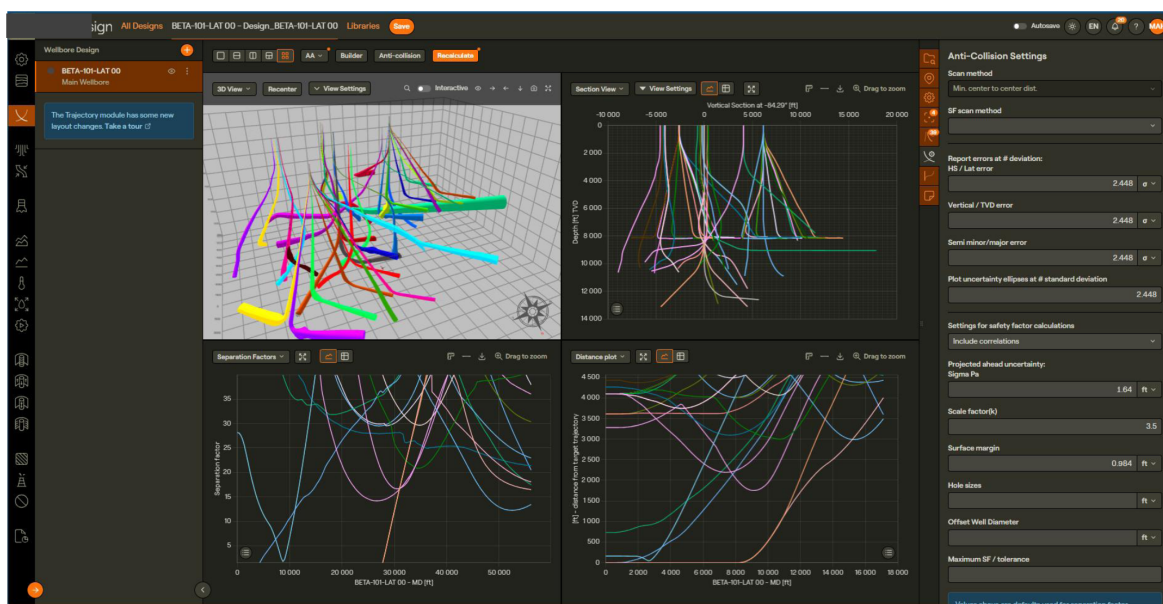


Figure 3—Well Design Software Graphical User Interface (GUI)

None of the software services requires specific cloud platforms. The software tool is devolved as Single Page Application (SPA), it does not use any proprietary cloud APIs. This means software tool services can run on any cloud; the only requirement is to spin up a Linux machine on it which absolutely all cloud platforms offer.

Single Page Application

In today's world, web applications are gradually replacing the old desktop applications as they are more convenient to use, they are easy to update, and they are not bound to any one device. Today's Petroleum Engineers are using these types of applications every day (for instance: Gmail, Google Maps, or Facebook), and they want to have similar software experience also in their work tasks.

A single-page application is a type of application that works inside a web browser and does not require page reloading during its usage. SPAs are all about enhancing User Experience by trying to imitate a natural environment in the browser with no page reloads, no extra wait time. It is just one web page that we visit which then loads all other content using JavaScript.

Single-page sites help keep the user in one space where content is presented in a simple, easy and workable fashion. SPA requests the markup and data independently and renders pages straight in the browser. This can be done using advanced JavaScript frameworks like AngularJS, ReactJS, Ember.js, Meteor.js, Knockout.js.

Application Programming Interface (API)

API stands for Application Programming Interface. In the context of APIs, the word Application refers to any software with a distinct function. Interface can be thought of as a contract of service between two applications. This contract defines how the two communicate with each other using requests and responses. Their API documentation contains information on how developers are to structure those requests and responses. APIs are mechanisms that enable two software components to communicate with each other using a set of definitions and protocols. For example, the weather bureau's software system contains daily weather data. The weather app on your phone "talks" to this system via APIs and shows you daily weather updates on your phone.

APIs can function in various ways depending on their purpose and the time of their development. Among the most popular and versatile types in use on the web today are REST APIs (Representational State Transfer). In REST APIs, the client sends requests to the server as data. The server uses this client input to start internal functions and returns output data back to the client. Let's look at REST APIs in more detail below.

The subject cloud-based Well Design tool uses open REST APIs to empower users to work towards achieving full automation. These APIs act as connectors, allowing data to flow seamlessly and enabling teams to all have access to the same information and work more collaboratively. The tool APIs are all defined as REST services taking JSON payload protected with JWT Bearer Tokens. We strive to follow the REST design pattern, which uses the HTTP protocol to work directly with resources. RESTful applications commonly make use of the following HTTP verbs for describing how resources should be managed:

- POST - For creating a resource (Create)
- GET - For fetching a resource (Read)
- PUT or PATCH - For replacing or updating a resource (Update)
- DELETE - for deleting a resource (Delete)

Automatic Well Design Workflow

A set of rules and simulation parameters is defined within the cloud-based software tool, enabling the end user to perform the following design tasks automatically with just a few mouse clicks:

- Multiple Trajectories Automatic Generation,
- Casing & Tubing Design,
- Analysis for Drilling Mechanics Torque & Drag/ Hydraulics,
- Well Operations Temperature Simulations,
- Simulations for Blow Out Worst-Case Discharge & Dynamic Kill.

To illustrate the above automated tasks, the following section provides a more detailed description of Automated Multiple-Well Trajectory Generation and Automated Casing Design.

Automatic Well Trajectory Workflow

In this workflow, the software algorithm will automatically generate the shortest directional trajectory to pre-defined geological target within the given design limits (depth of kick off, and max. allowed dog leg severity). The directional planning method applied in the software algorithm is "Optimum alignment". To demonstrate this automatic trajectory generations workflow, we are going to use the following example of a hypothetical Field Development Plan, see [Figure 4](#):

- Ten (10) geological target pairs, each pair with an Entry target and a Final Target
- All Wells start from the same surface location (a drilling application of PAD/Platform Multi-wells)

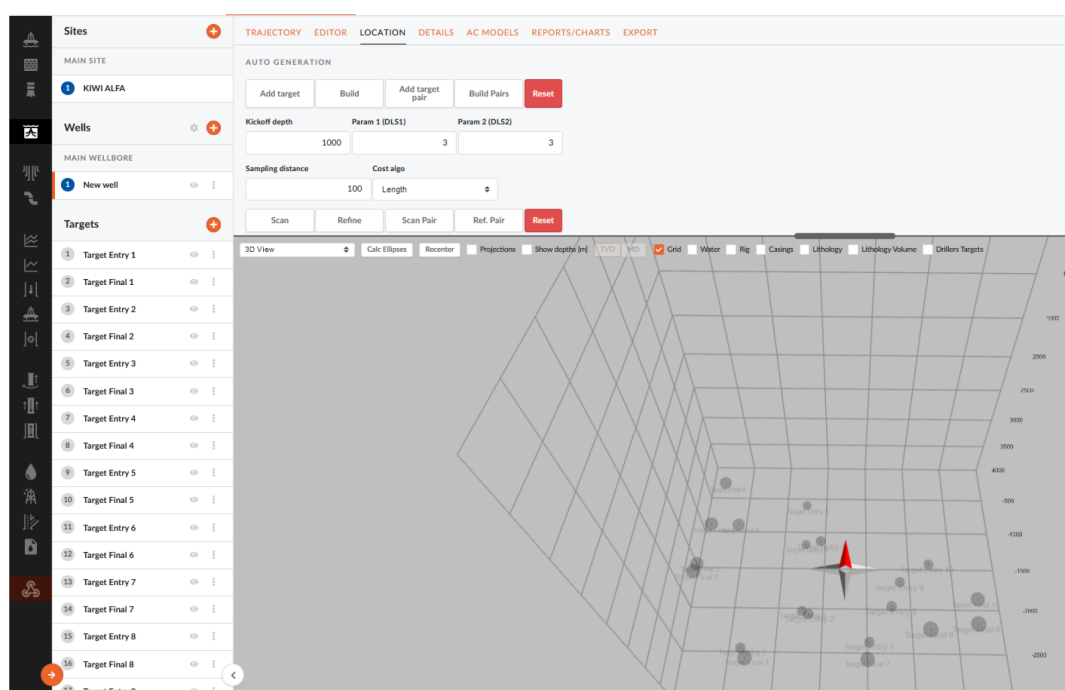


Figure 4—PAD Multi-wells Filed Development Example, Ten (10) geological target pairs

With just one mouse click, the software will automatically generate the well trajectories from the given PAD/Platform surface location.

Software can also run hundreds of iterations to optimize the PAD/Platform surface location while hitting all geological targets with shortest possible well trajectories. See [Figure 5 & 6](#)

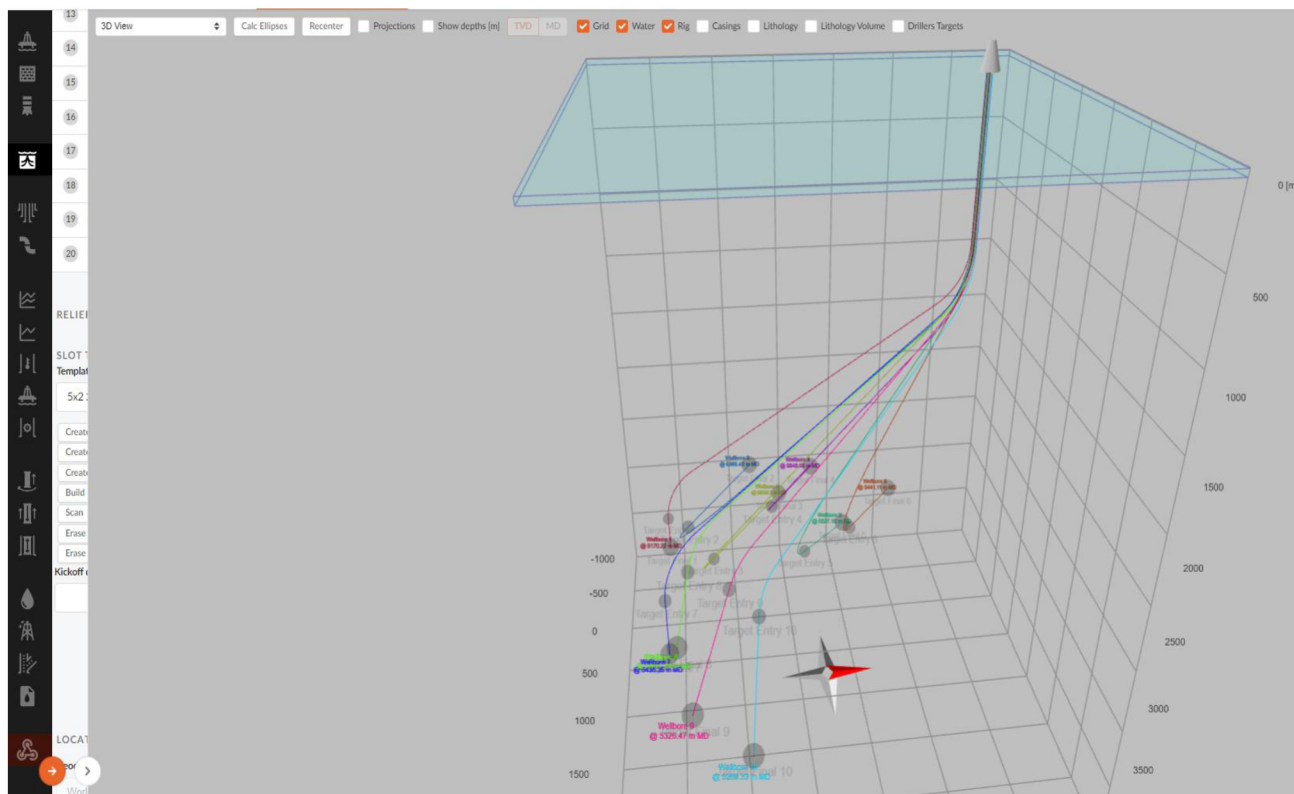


Figure 5—Example of Auto-generated well trajectories

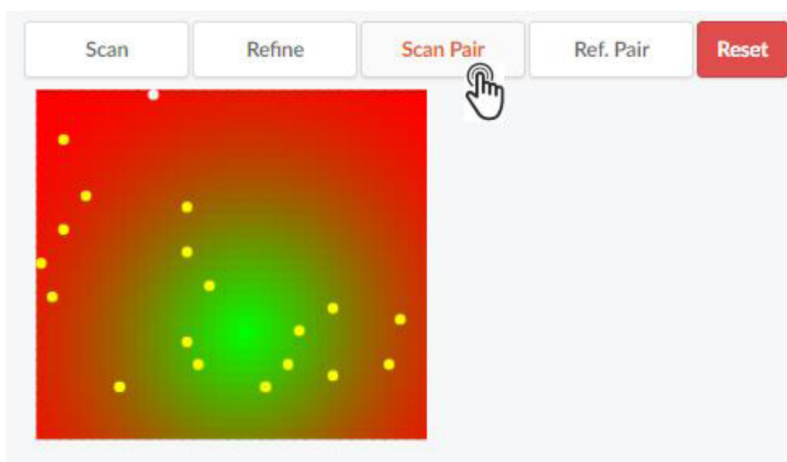


Figure 6—Example of heat-map for optimum surface location selection

After refining the surface location, software algorithm will re-calculate well trajectories (see [Figure 10](#)). In this example, there is almost a 30% reduction in the combined total drilling meterage of the 10 wells.

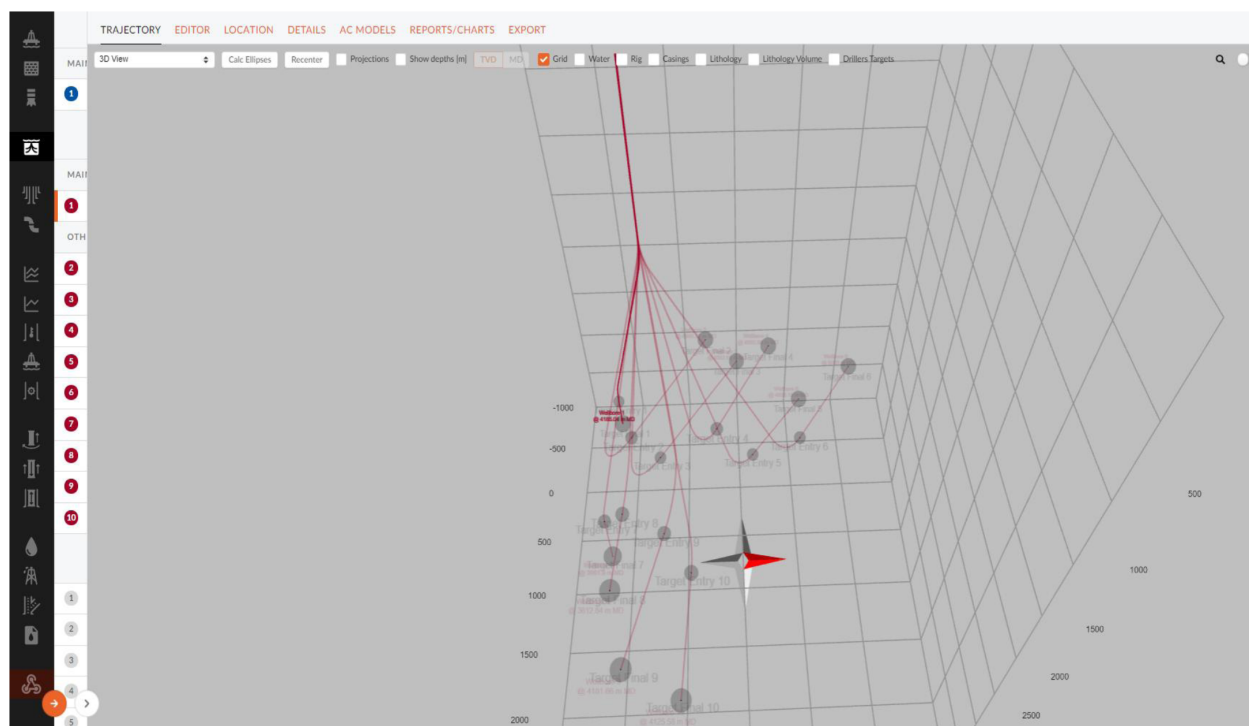


Figure 7—Example of refined trajectories after surface location optimization.

Automatic Casing Design Workflow

This workflow allows for automatic casing and tubing design based on a pre-defined "Rule Set" which includes the criteria for selection casing setting depth, casing design loads (Burst, Collapse, and Axial Loads), Design Factors, and tubular availability (string priority). A "Rule Set" can be created for a field (drilling area) or well architecture (large-bore, reduced-bore, lean profile, ..etc). Then, any user can call the "Rule Set" from the software library to apply it to any design project, and the software will automatically execute full casing design check, generate all output charts & tables, and calculate resultant safety factors.

Following is an example of automatic casing design "Rule Set". See [Figure 8](#).

Once Automatic Casing Design is selected, the "Rule Set" tab appears and desired rule set is chosen from the software library, and the validation of the chosen rule set is visualized. Then, at a mere click of a mouse, software will automatically choose the casing setting depths, mud weights, and calculate kick tolerances. Then it will simulate different casing loads against different pipes weights & grades and calculates final safety factors.

GENERAL
SETTING DEPTH
LOAD CASES
STRING PRIORITY

STRING FUNCTION PRIORITY

Function	String type	Setting Criteria	Cementing Criteria	Mud type	+
Conductor	Casing	90m depth below mudline	Cement to mudline	Seawater	
Surface	Casing	35m below Formation 2	Cement to mudline	Default WBM	—
Intermediate	Casing	50m below Formation 3	Cement to mudline	Light OBM	—
Intermediate	Casing	100m below Formation 4	Cement to mudline	Light OBM	—
Production	Casing	100m below Formation 5	Cement to mudline	Medium OBM	—
Intermediate	Casing	750m below Formation 6	Cement to mudline	Medium SBM	—
Production	Casing	10m below Formation 7	Cement to mudline	Medium SBM	—
Production	Liner	200m below Formation 8	Cement to liner top	Heavy OBM	—
Production	Liner	Kick Tolerance Table	Cement to liner top	Heavy OBM	—

Mud strategy

☐ Maximum pore pressure & margin
☒ Middle value Pore/Frac

Error margins (Margins will be used in calculations)

Pore Pressure
0.05
sg
Fracture Pressure
0.05
sg

GENERAL
SETTING DEPTH
LOAD CASES
STRING PRIORITY

CONDUCTOR - CASING

BURST

☐ Displacement To Gas
☐ Frac@Shoe/GG Above
☐ Drill Ahead (Burst)
☐ Gas Kick Profile
☒ Pressure Test
☐ Green Cement Pressure Test (Burst)
☐ Lost Returns w/Water
☐ Surface Protection (BOP)
☐ Gas Over Mud Ratio

COLLAPSE

☐ Drill Ahead (Collapse)
☒ Casing Evacuation
☐ Lost Returns With Mud Drop
☒ Cementing
☐ WCST
☐ Riser Margin
☐ Negative Pressure Test

INSTALLATION LOAD CASES

☐ Overpull
☒ Running in hole
☐ Pre-Cement Static Load

SURFACE - CASING

BURST

☐ Displacement To Gas
☐ Frac@Shoe/GG Above
☐ Drill Ahead (Burst)
☐ Gas Kick Profile
☒ Pressure Test
☐ P6A Pressure Test
☐ Green Cement Pressure Test (Burst)
☐ Lost Returns w/Water
☐ Surface Protection (BOP)
☐ Gas Over Mud Ratio

COLLAPSE

☐ Drill Ahead (Collapse)
☐ Casing Evacuation
☐ Lost Returns With Mud Drop
☒ Cementing
☐ WCST
☐ Riser Margin
☐ Negative Pressure Test

INSTALLATION LOAD CASES

☐ Overpull
☒ Running in hole
☐ Pre-Cement Static Load

INTERMEDIATE - CASING

BURST

☐ Displacement To Gas
☐ Frac@Shoe/GG Above
☐ Drill Ahead (Burst)
☐ Gas Kick Profile
☒ Pressure Test
☐ P6A Pressure Test
☐ Green Cement Pressure Test (Burst)
☐ Lost Returns w/Water
☐ Surface Protection (BOP)
☐ Gas Over Mud Ratio

COLLAPSE

☒ Drill Ahead (Collapse)
☐ Casing Evacuation
☐ Lost Returns With Mud Drop
☒ Cementing
☐ WCST
☐ Riser Margin
☐ Negative Pressure Test

INSTALLATION LOAD CASES

☐ Overpull
☒ Running in hole
☐ Pre-Cement Static Load

INTERMEDIATE - CASING

BURST

☐ Displacement To Gas
☐ Frac@Shoe/GG Above
☐ Drill Ahead (Burst)
☐ Gas Kick Profile
☒ Pressure Test
☐ P6A Pressure Test
☐ Green Cement Pressure Test (Burst)
☐ Lost Returns w/Water
☐ Surface Protection (BOP)
☐ Gas Over Mud Ratio

COLLAPSE

☒ Drill Ahead (Collapse)
☐ Casing Evacuation
☐ Lost Returns With Mud Drop
☐ Cementing
☐ WCST
☐ Riser Margin
☐ Negative Pressure Test

INSTALLATION LOAD CASES

☒ Overpull
☒ Running in hole
☐ Pre-Cement Static Load

PRODUCTION - CASING

BURST

☐ Displacement To Gas
☐ Frac@Shoe/GG Above
☐ Drill Ahead (Burst)
☐ Gas Kick Profile
☒ Pressure Test
☐ P6A Pressure Test
☐ Green Cement Pressure Test (Burst)
☐ Tubing Leak
☐ Lost Returns w/Water
☐ Surface Protection (BOP)
☐ Stimulation Surface Leak
☐ Injection Down Casing
☐ Perforating Gun Pressure
☐ Gas Over Mud Ratio

COLLAPSE

☒ Drill Ahead (Collapse)
☐ Casing Evacuation
☐ Full Evacuation
☐ Gas Migration
☐ Lost Returns With Mud Drop
☐ Cementing
☐ WCST
☐ Above/Below Packer

Figure 8—Example of Automatic Casing Design "Rule Set"

For illustration see the following Before/ After example. See Figure 9 & 10.

BEFORE:

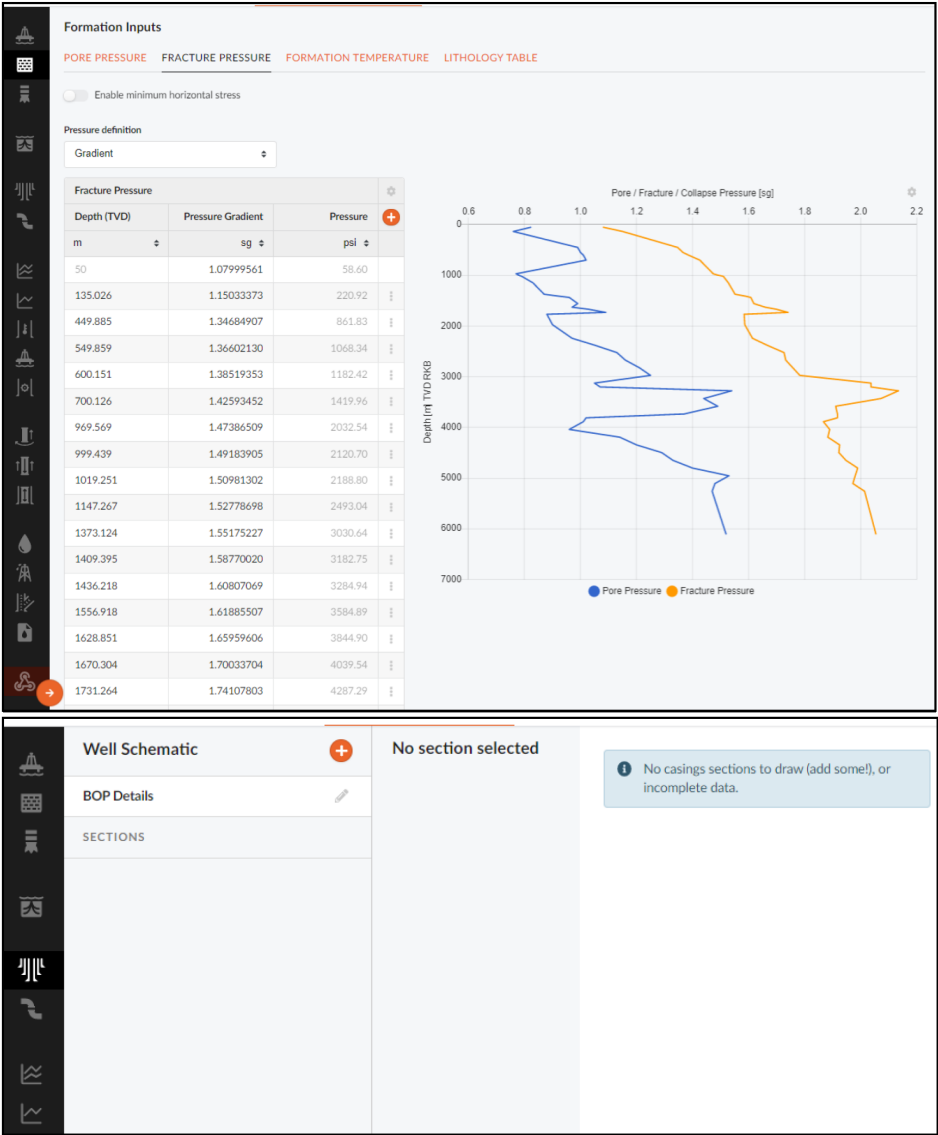


Figure 9—BEFORE Example: empty well design case before applying Automatic Casing Design, no casing profile or design loads.

AFTER:



Figure 10—AFTER Example: results of Automatic Casing Design

Carbon Capture and Storage (CCS) Well Design Using Cloud Software

In addition to standard well design & drilling engineering calculations tasks, the software tool also enables design of CO₂ CCS Wells. To ensure safe and environmentally sound CCS Well Delivery, a detailed well design should be performed with the following criteria: 1) Dynamic thermal wellbore simulation with full capability to model liquid (Supercritical) CO₂ injection. 2) Multi-string casing design with the option to add and remove fluids in any annuli. 3) Ability to simulate liquid CO₂ blowout and kill operations.

To address CO₂ CCS well design requirements, the cloud-based software tool is built on top of an extensive fluids library that, amongst others, contains predefined CO₂ fluids, both pure and impure. The

library also supports addition of custom fluid files, either by uploading an OLGA format tab file or by defining a custom fluid composition. Currently, the hydrocarbon equation of state is applied when defining a CO₂ fluid in the software hydrocarbon library. When using the thermal simulator in the tubing design module, CO₂ can be selected as the applied fluid for production and injection simulations.

The chained simulation functionality further allows these results to be used as the initial condition when simulating waiting or shut-in operations. The thermal simulator has full capability to model liquid CO₂, including the full phase envelope of CO₂ and the Joule Thompson effect from gas expansion and Multistring analysis includes annular fluid expansion and annular pressure build up. In addition to negative fluid expansion, the software tool supports addition and removal of volume from any annulus, both before and after a thermal change. See Figure 11. Further CO₂ modeling enhancement in the cloud software tool is currently ongoing for the implementation of the Span & Wagner model, a full equation of state specifically for CO₂, and dual phase calculations

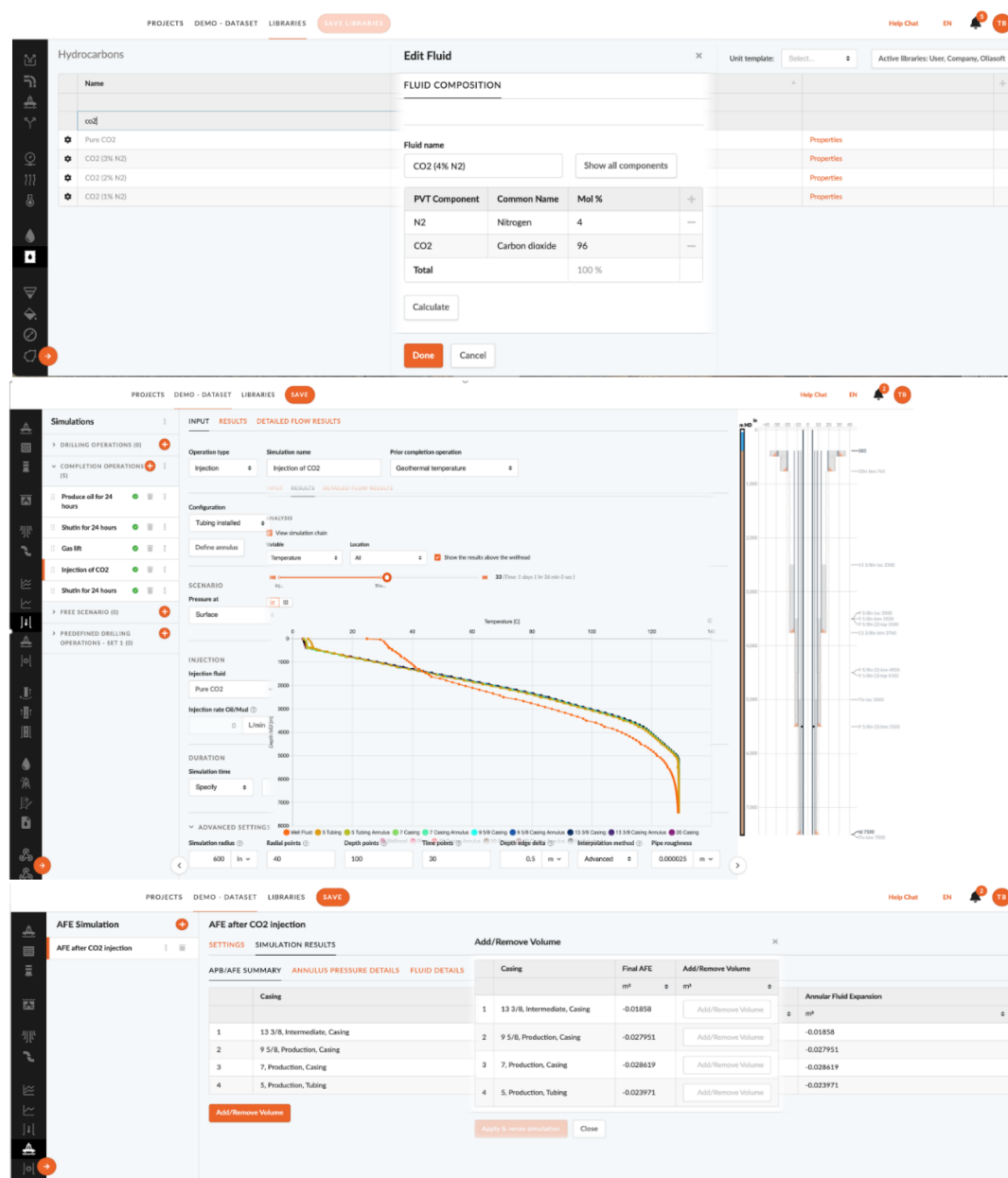


Figure 11—Example CCS Well Design Using Cloud Software Tool

Improved Workflow

Below is a summary of the differences between the improved workflow using Ruleset-driven Automated design versus the Legacy software workflow:

Table 2—Workflow comparison of Ruleset-driven Automated design versus Legacy software workflow

#	Item	Ruleset-driven Automated Workflow	Legacy Software Workflow
1	Work Time on Software	Potential saving 70% or more of the work time (manhours) on software. Design performed in few minutes instead of many hours/ days	Time consuming require more workdays/weeks to perform a full PAD/Platform Multi-wells design
2	Input Collection for Geological & Reservoir Data (targets, Lithology, Pressures, ..etc)	enables API machine to machine communication with Geology & Reservoir software. Data can be communicated to Well Design cloud-based tool automatically	Manual input, several emails exchange between Geology & Drilling teams. Data must be exported from Geology & Reservoir software (in Excel, Text, or another format) then imported to legacy drilling software
3	Drillability Study (multiple Well Trajectories design from single template)	Automated Trajectories Design and optimum surface location selection integrated in same Well Design cloud-based tool	Engineers design each well trajectory separately, and iterate designs many times with different template surface location positions. Engineers use separate software for Trajectory design
4	Casing & Tubing Design	Automated "Rule Based" casing design integrated in same Well Design cloud-based	Manual (conventional) casing design process using legacy software Engineers use separate software for Casing design, and another separate software temperature simulations
5	Wellbore Engineering/ Drilling Mechanics: (Torque & Drag Analysis, Hydraulics Analysis, Surge & Swab)	Automated "Rule Based" drilling mechanics simulations integrated in same Well Design cloud-based	Engineers use separate software for drilling mechanics
6	Blow-out & Kill simulations	performs stochastic blowout and dynamic kill calculations integrated in same Well Design cloud-based	Not available. Drilling Engineers outsource this work to external consultancy firms to perform Blow-out & Kill studies
7	Carbon Capture & Storage (CCS) CO2 Wells simulations	Supported, tool can simulate liquid CO2 injection	Not available.
8	User Experience	Web-based cloud application, user-friendly interface. Intuitive & Circular workflow between design modules	Classical desktop old interface, serval processes are bit complicated to use by average user. Workflow process split between different separate desktop software
9	Software Architecture	Modern, cloud-native software architecture utilizes a distributed microservices approach to build independent, scalable, and resilient applications that take full advantage of cloud computing. Enabler for Digital Transformation	Monolithic, Desktop software archticture. All software components (UI, business logic, data access) are tightly coupled and run as one large unit

Conclusion

Well Construction Automation is gradually becoming more prominent in oil & gas industry. It encompasses the application of digital technology in all aspects of well drilling and completion (i.e., automatic well design, digitalization of downhole tools & surface equipment, remote monitoring, real time data transmission, and robotic rig systems). In today's Industry 4.0 transformation world, the cloud-based software can automate Oil & Gas wells design. Future software development should be made to produce fully autonomous Oil & Gas well designs while maintaining industry safety & well integrity standards

The software's modular architecture allows seamless data transfer across all design stages. Interconnected calculation engines ensure automated, iterative computations without the need for redundant manual data

entry or validation, enabling real-time updates to well designs based on rule logic. For any new PAD/Platform development, the user can initiate the automated workflow via rule-based templates.

The software autonomously executes the full suite of simulations and generates engineering designs across all wells in the PAD/Platform. This approach achieved significant reductions in design cycle time while improving quality control and standardization. This approach demonstrates the viability of fully automated design of PAD/Platform Multi-wells, enabling complex planning tasks to be executed within minutes through intelligent, cloud-native digital tools.

Nomenclature

API	= Application Programing Interface
CFD	= computational fluid dynamics
CSPs	= Cloud Service Providers
CCS	= Carbon capture and storage
HTTP	= Hypertext Transfer Protocol
IaaS	= Infrastructure as a Service
JSON	= JavaScript Object Notation
JWT	= JSON Web Token
PaaS	= Platform as a Service
REST	= Representational state transfer software architectural style
SaaS	= Software as a Service
SPA	= Single Page Application
SQL	= Structured Query Language
SSL	= Secure Sockets Layer

References

1. de Wardt, John P, Macpherson, John D, Zamora, Mario, Dow, Blaine, Hbaieb, Slim, Macmillan, Robin A, Laing, Moray L, DiFiore, Amanda M, Inabinett, Calvin E, and Mark W Anderson. 2015. Drilling Systems Automation Roadmap - The Means to Accelerate Adoption. Paper SPE-173010-MS presented at the SPE/IADC Drilling Conference and Exhibition, London, England, UK, March 2015
2. Mathieson, Derek, Meehan, D. Nathan, and Jeff Potts. 2019. The End of Petroleum Engineering as We Know It. Paper presented at the SPE Middle East Oil and Gas Show and Conference, Manama, Bahrain, March 2019.
3. Carpenter, Chris. "The Cloud Offers Opportunity for Oil and Gas Cybersecurity." *J Pet Technol* **71** (2019): 72–73.
4. Benefits of Cloud Computing, *IBM* website <https://www.ibm.com/ae-en/cloud/learn/benefits-of-cloud-computing>.
5. Brechan, Bjorn, and Sigbjorn Sangesland. "Digital Well Planning, Well Construction and Life Cycle Well Integrity: The Role of Digital Experience." Paper presented at the SPE Norway One Day Seminar, Bergen, Norway, May 2019
6. Caso, Carlo, Bennett, Peter, Isaachsen, Jan, Ask, Kjell Kristian, Stray, Thorkild, Andresen, Per Arild, and Carl Fredrik Straumsheim. "Toward the Redefinition of Drilling Plan and Execution Via a Digital Drilling Ecosystem." Paper presented at the IADC/SPE International Drilling Conference and Exhibition, Galveston, Texas, USA, March 2020.
7. W. Eustes et al., 'The evolution of automation in drilling,' in *SPE Annual Technical Conference and Exhibition, Society of Petroleum Engineers*, 2007, pp. 1–5.